AI 3000: Reinforcement Learning
Practice Problem Set 2

1 Value Iteration in a 2-State MDP

Consider an infinite-horizon MDP with state space $\mathcal{S} = \{s_1, s_2\}$, action space $\mathcal{A} = \{a_1, a_2\}$, and discount factor $\gamma = 0.9$. The transition probabilities $P(s'|s, a)$ and reward $r(s, a, s')$ are given in the table below:

State (s)	Action (a)	$P(s_1\|s, a)$	$r(s, a, s_1)$	$r(s, a, s_2)$
s_1	a_1	0.8	5	10
s_1	a_2	0.5	12	8
s_2	a_1	0.2	15	0
s_2	a_2	0.5	8	4

1.1. Compute the expected one-step reward $r(s, a)$ for all state-action pairs $(s, a) \in \mathcal{S} \times \mathcal{A}$.

1.2. What is the optimal value function $v^*_{\infty,\lambda}(s)$ and optimal policy π^*?

1.3. Starting from initial value estimates $v_0 = (0, 0)$, perform value iteration for 3 steps for the following algorithms:

(a) Synchronous value iteration with span criterion

(b) Synchronous value iteration with sup-norm criterion

(c) Gauss-Seidel value iteration (updating state s_1 first)

1.4. Calculate the error in the sup-norm $\|v - v^*_{\infty,\lambda}\|_\infty$ for the value function v returned by the each algorithm.

2 Contraction mappings

Given any randomized decision rule $\phi\colon \mathcal{S} \to \Delta(\mathcal{A})$, show that $L_\phi\colon \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ given by

$$L_\phi v = r_\phi + \lambda P_\phi v$$

is a contraction mapping on $\mathbb{R}^{|\mathcal{S}|}$.

Subsequently, prove the monotonic property of L_ϕ analogous to the maximum return operator L, i.e. Theorem 5.7 from Lecture 8 (reproduced below).

2.1. If there exists $v \in \mathbb{R}^{|\mathcal{S}|}$ such that $v \geq L_\phi v$, then $v \geq v^\phi_{\infty,\lambda}$.

2.2. If there exists $v \in \mathbb{R}^{|\mathcal{S}|}$ such that $v \leq L_\phi v$, then $v \leq v^\phi_{\infty,\lambda}$.

2.3. If there exists $v \in \mathbb{R}^{|\mathcal{S}|}$ such that $v = L_\phi v$, then $v = v^\phi_{\infty,\lambda}$.

(All vector inequalities are component-wise).